Carboxylic Acid

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Carboxylic Acid

INTRODUCTION

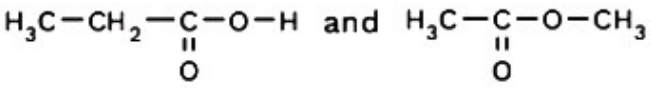
(a) Organic compounds in which –COOH group is present are named "Carboxylic Acid".

(b) These compounds are also called fatty acids because their higher members like Palmitic acid or stearic acid were usually obtained from body/Animal fats.

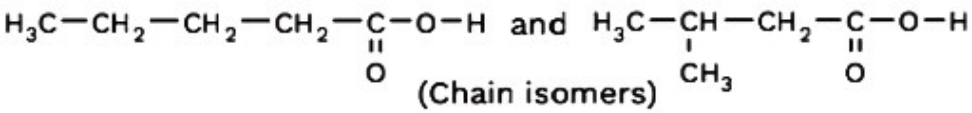
(c) The term 'Carboxylic' was first proposed by inventor 'Bayer'.

(d) Their general formula is $CnH_{2n}O_2$.

Hybridisation state of carboxylic carbon is sp^2. In carboxylic acids, both oxygen-carbon bond distances are equal due to resonance. It is 1.28 A°. On the base of number of –COOH group, they are called mono, di or tri carboxylic acid. Acids shows chain and functional isomerism. They show functional isomerism with ester.



(Functional isomers)

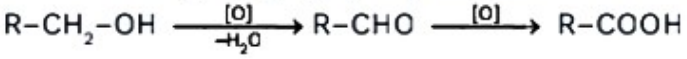


(Chain isomers)

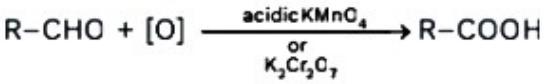
METHODS OF PREPARATION

(a) By oxidation of primary alcohol:

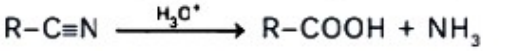
Acidic $KMnO_4$ or $K_2Cr_2O_7$ may be cast-off in the below kind of reaction.



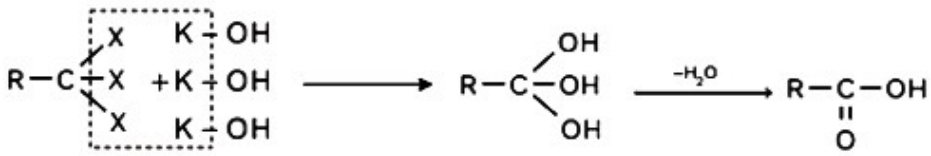
(b) By oxidation of Aldehyde and Ketone:



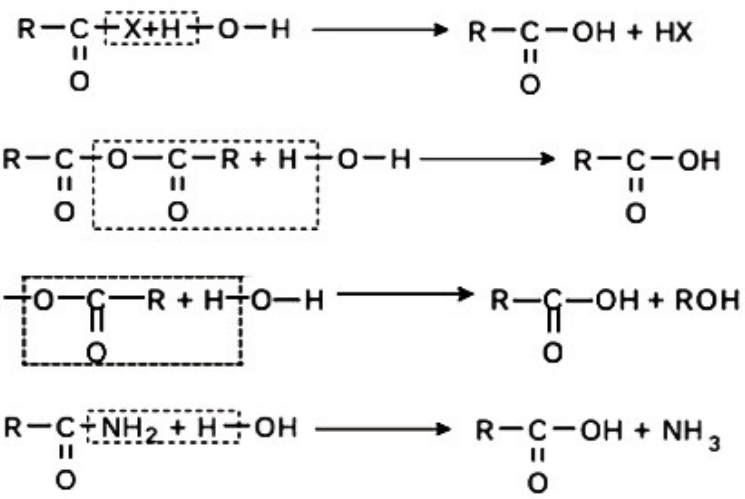
(C) From Alkyl Cyanide: Cyanides gives acid on complete hydrolysis.



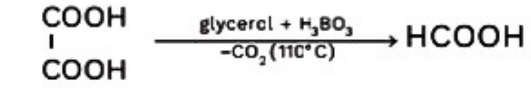
(d) From Trihaloalkane : 1, 1, 1- Trihaloalkane reacts with aq. KOH and forms respective acid.



(e) From Acid derivative (By hydrolysis) :

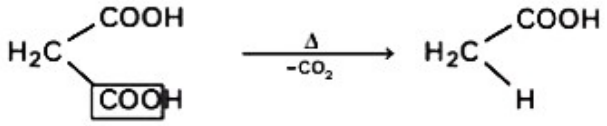


(f) From Dicarboxylic Acid (Decarboxylation)



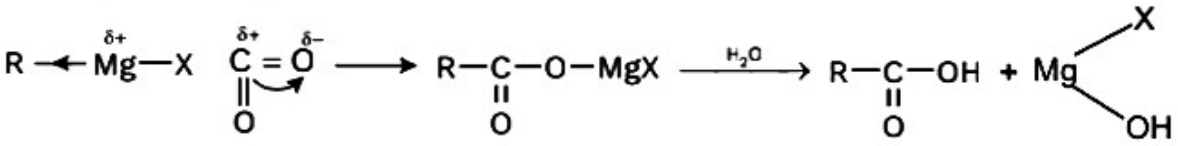
Oxalic acid

Note : The above reaction is lab technique to make formic acid.

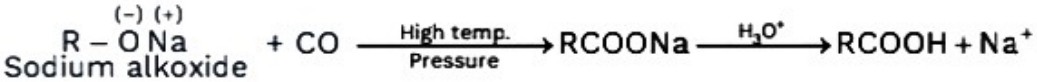


malonic acid ethanoic acid

(g) From Grignard Reagent:

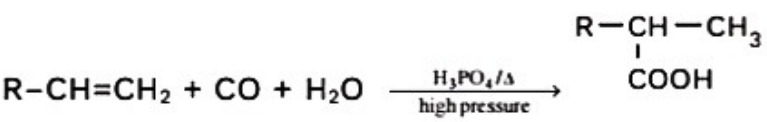


(h) From Carbonylation of sodium alkoxide:



Sodium alkoxide

(i) From Alkenes: (CO, H_2O)



(j) From Arndt-Eistert process:

This reaction is used in ascending sequence. In this reaction alkanoyl halide treated with diazomethane and acid is produced on hydrolysis.

$$CH_3COCl + CH_2N_2 \xrightarrow{-HCl} CH_3COCHN_2 \xrightarrow[-N_2]{Ag_2O} CH_3-CH=C=O \xrightarrow{H_2O^+} CH_3-CH_2-COOH$$

(k) Manufacture of formic acid only :

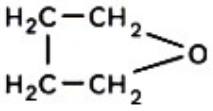
(a) $NaOH + CO \xrightarrow[6-10\ atom.]{210^\circ} HCOONa$

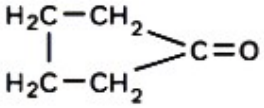
$$HCOONa \xrightarrow{H_2SO_4} HCOOH$$

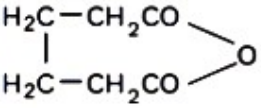
(b) $CO + H_2O \xrightarrow[pressure/BF_3]{high} HCOOH$

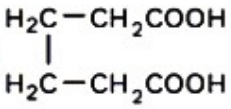
(c) $CH_3OH + CO \xrightarrow[high pressure]{200^\circ} HCOOCH_3 \xrightarrow{dil.H_2SO_4} HCOOH + CH_3OH$

Q25 Which of the subsequent products is formed when adipic acid is intense

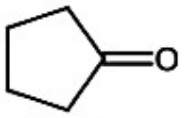
(A) 

(B) 

(C) 

(D) 

Sol: (B)

$$HCOO(CH_2)_4\ COOH \xrightarrow[\Delta]{300^\circ C}$$ 

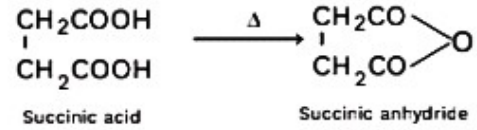
cyclopentanone

Q26 **Malonic acid and succinic acids are notable by**

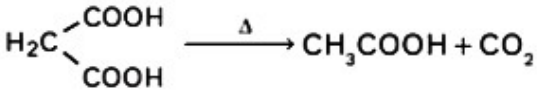
(A) Heating **(B) $NaHCO_3$**

(C) Both A & B **(D) None of these**

Sol: **(A)**

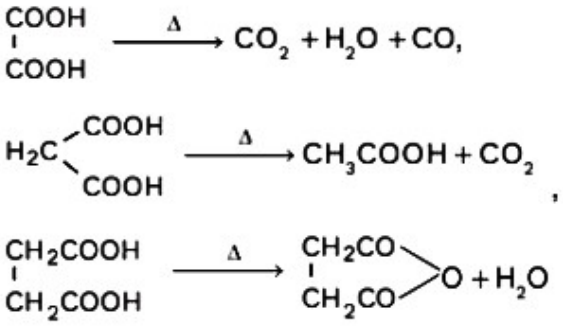


Succinic acid Succinic anhydride



Malonic acid

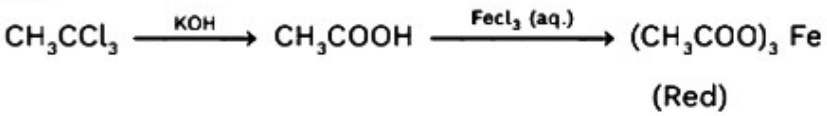
Two carboxylic acid group on one carbon atom gives CO_2 on heating.
Two carboxylic group an adjacent carbon atoms lose H_2O towards give an hydride on heating



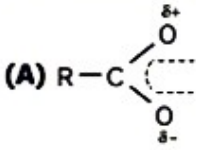
Q27 **The hydrolysis of 1, 1, 1-trichloro derivative (A) of alkane provide a molecule (B) on basic hydrolysis which produces red color with aq. $FeCl_3$. The molecule (A) is**

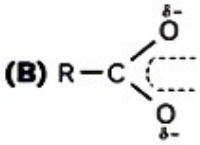
(A) $CH_3CH_2CCl_3$ **(B) CH_3CCl_3** **(C) $CHCl_3$** **(D) None**

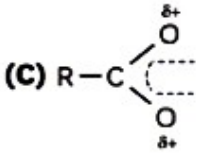
Sol: **(B)**



Q28 **Which of the subsequent is the best illustration of the structure of the carboxylate ion?**

(A) 

(B) 

(C) 

(D) None of these

Sol: **(B)**
Both C–O bonds are undistinguishable and each O holds partial charges.

PHYSICAL PROPERTIES

(a) Carboxylic acids are colourless with specific smell liquid. They are in liquid state upto 9 carbon and above 9 carbon they are solids.

(b) Lower carboxylic acids soluble in water. It is due to H–bonding. Higher carboxylic acids are insoluble in water due to increase in covalent character.

(c) Their melting point and boiling point $\propto$ molecular mass

$$\propto \frac{1}{\text{No. of branches}}.$$

(d) Boiling point of acids are moderately higher than equivalent esters due to H–bonding. For example, methyl formate ($HCOOCH_3$) takes a boiling point 57°C while acetic acid has a boiling point 118°C. Boiling point of formic acid 100.5°C. For elimination of water from formic acid we can use reagent $PbCO_3$ (lead carbonate) and H_2S.

(e) Formic acid is used in leather industries, also known as corrosive.

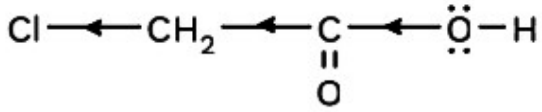
(f) 8–10% acetic acid is called **'Vinegar'**

(g) At 16–20°C pure acetic acid converts into ice like solid called glacial acetic acid.

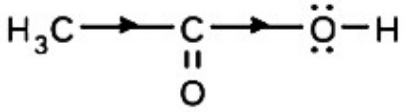
(h) In industries, acetic acid is prepared by oxidation of ethanol and oxidation occurs by aceto bacter acetic or micoderma acetic bacterias. The process is called quick vinegar process.

Acidic strength

(j) Acidic nature of an acid can be increased by the count of –I groups in place of α-hydrogen atoms. Acidic strength can be decreased by adding of + I groups.



More acidic



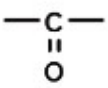
Less acidic

Chemical Characterictics

Chemical reaction of carboxylic acids are catogerised in five different class:

(1) Reaction of Hydrogen Atom of Carboxylic Acid.

(2) Reaction of –OH group of Carboxylic Acid.

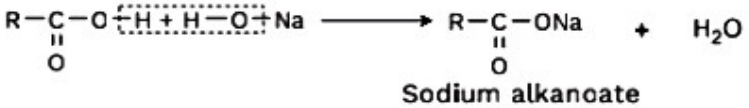
(3) Reaction of  group of Carboxylic Acid.

(4) Reaction of –COOH group of Carboxylic acid.

(5) Reaction of Alkyl group of Carboxylic acid.

1. Reaction Of Hydrogen Atom of Carboxylic Acid

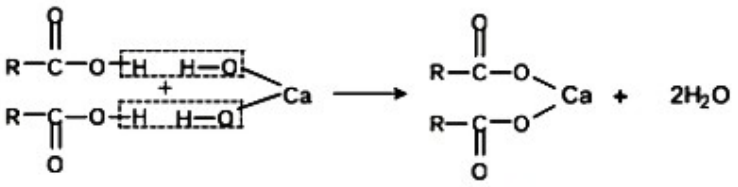
Carboxylic acid reacts with base and forms salt, thus showing acidic character.

i) 

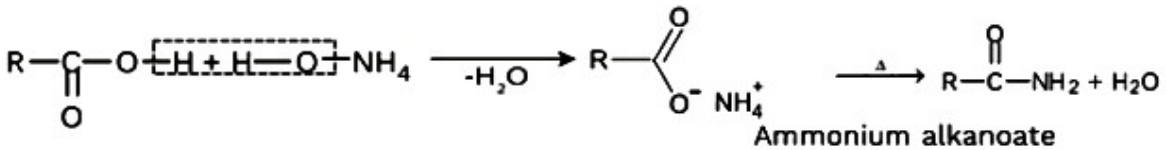
Sodium alkanoate

ii) 

Sliver alkanoate

iii) 

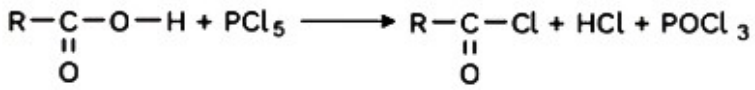
Calcium alkanoate

iv) 

Ammonium alkanoate

2. Reaction of hydroxy functional group of Carboxylic Acid :

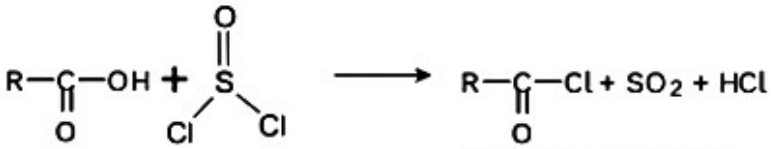
i) Reaction with PCl_5 or PCl_3 or $SOCl_2$:



Alkanoly chloride

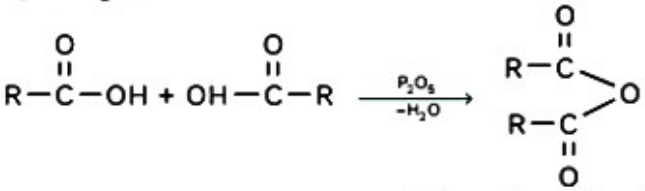


Alkanoly chloride



Alkanoly chloride

ii) Dehydration :

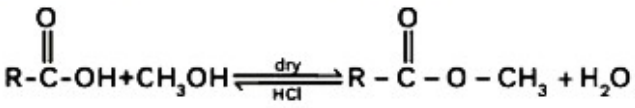


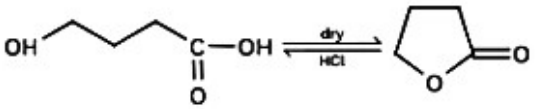
Alkanoic anhydride

iii) Reaction with Alcohol : SN_{AE} (nucleophilic substitution with addition and Elimination)

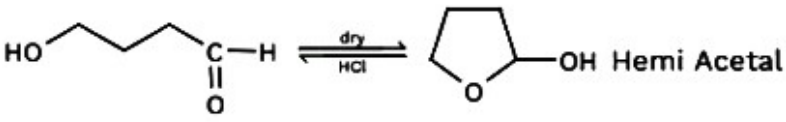
Formation of Ester with carboxylic acid and alcohol in dry acidic medium is called Esterification

Identify in which reaction ester is product?

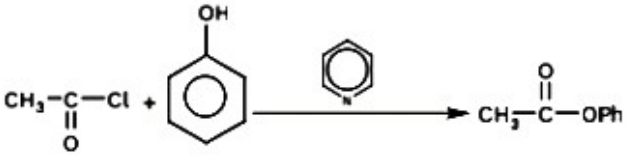
(i) 

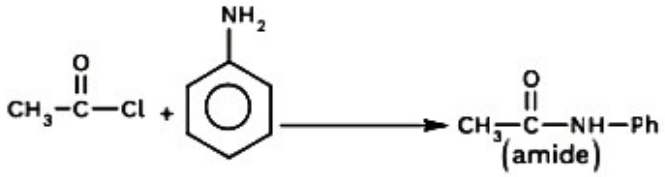
(ii) 

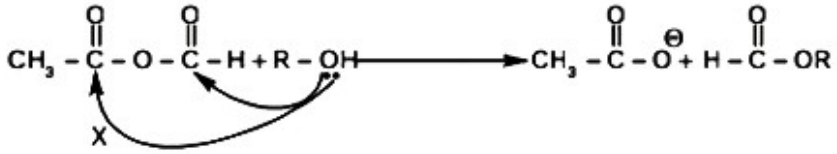
Lactone
Cyclic Ester

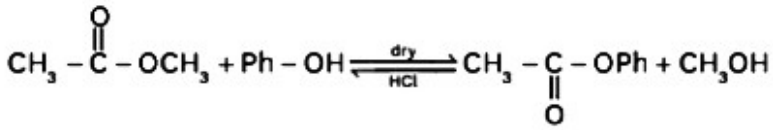
(iii) 

Hemi Acetal

(iv) 

(v) 

(vi) 

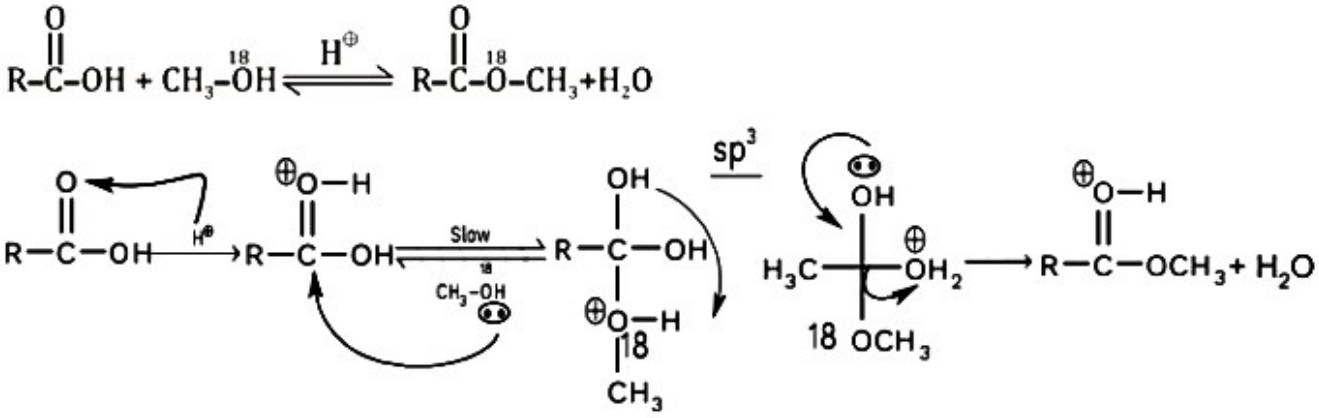
(vii) 

Esterification Reaction

It is a chemical reaction during the formation of the ester. Esterification is the chemical process that combines alcohol (ROH) and an organic acid (RCOOH) to form an ester (RCOOR) and water. This chemical reaction results in forming at least one product of ester through an esterification reaction between a carboxylic acid and an alcohol.

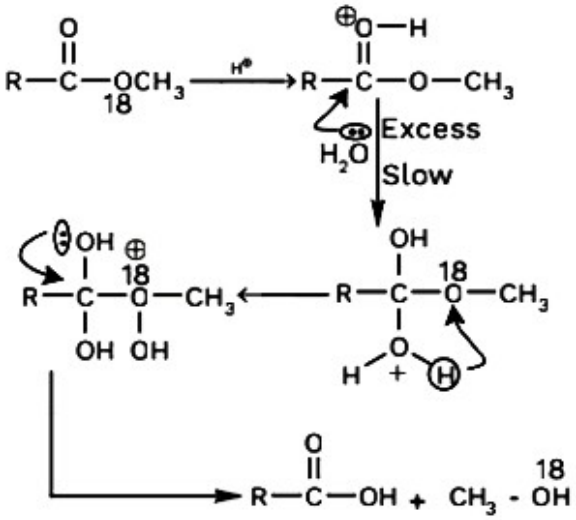
(a) Etherification is a laboratory test for carboxylic acid and alcohol because it gives fruity smell.

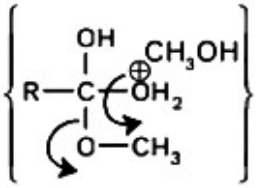
(b) Etherification is a reversible reaction in acidic medium to forward follow le-chatlier principle.



To forward reaction excess of alcohol or removal of H_2O is suggested

Backward



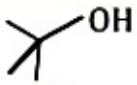


Note :

(i) If CH_3OH in excess Ester is major product

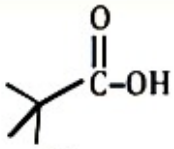
(ii) If H_2O in excess acid major product.

Rate of Esterification

Q29 **(i)** **CH_3OH** **C_2H_5OH** 

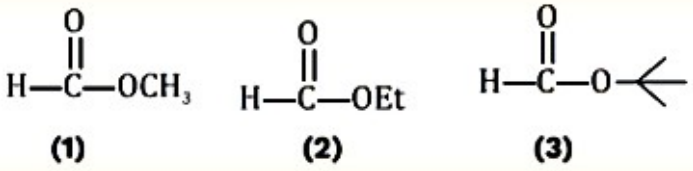
(1) **(2)** **(3)**

Sol: $\text{Rate of RxN} \propto \frac{1}{\text{Size of alcohol}} (1 > 2 > 3)$

Q29 **(ii)** **HCOOH** **CH_3COOH** 

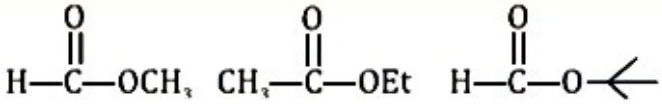
(1) **(2)** **(3)**

Sol: $\text{Rate of RxN} \propto \frac{1}{\text{Size of Acid}} (1 > 2 > 3)$

Q30 i) 

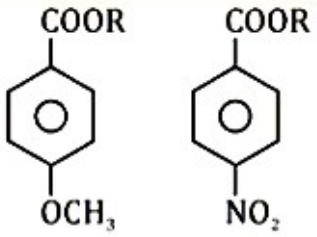
(1) (2) (3)

Sol: Smaller alcohol, higher rate of reaction $(1 > 2 > 3)$

Q30 ii) 

(1) (2) (3)

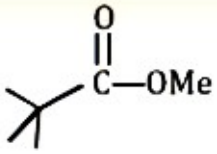
Sol: Aldehyde is more reactive than Ketone $(1 > 2 > 3)$

Q31 

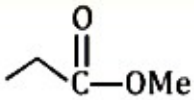
(1) (2)

Sol: $2 > 1(-NO_2 = -M, -OCH_3 = +M)$

Q32



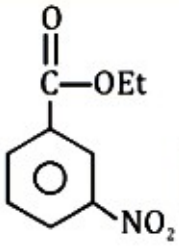
(1)

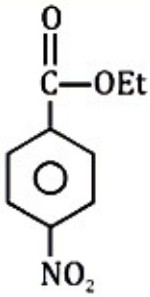


(2)

Sol: Size of carbonyl group $\propto \frac{1}{\text{R.O.R}}(2 > 1)$

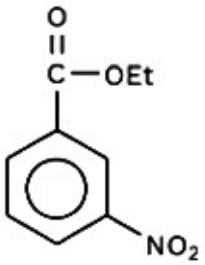
Q33

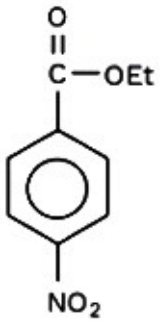




(1) (2)

Sol:



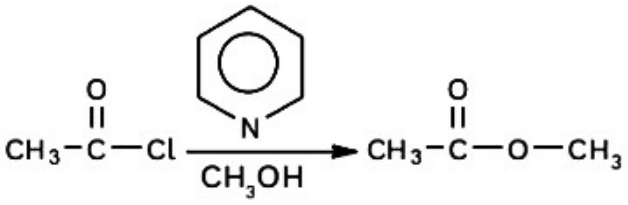


-I -M only

R.O.R will be (2 > 1)

Note:

Esterification can be irreversible in presence of acyl halide with pyridine which is called acylation and it is used for number of –OH group.

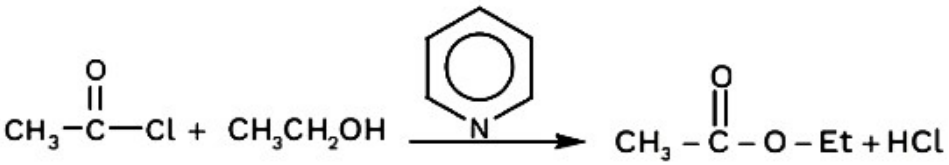


* Pyridine abstract $H^{\oplus}$ ion so reverse mechanism is not working.

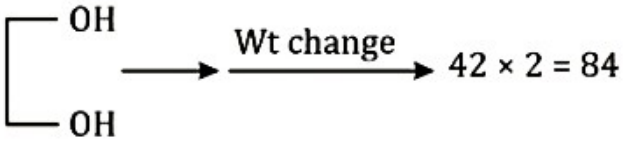
Note: 42 molecular weight change occurs in case of single -OH group.

Application :

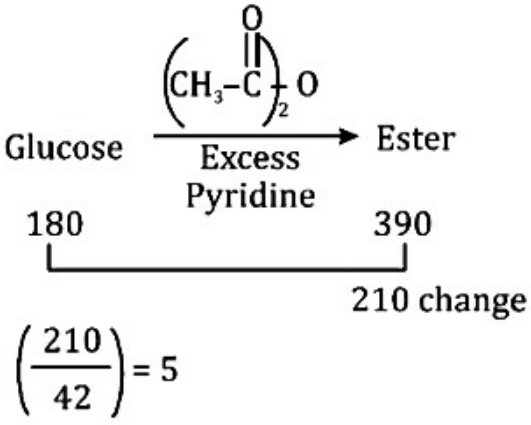
Ex-1



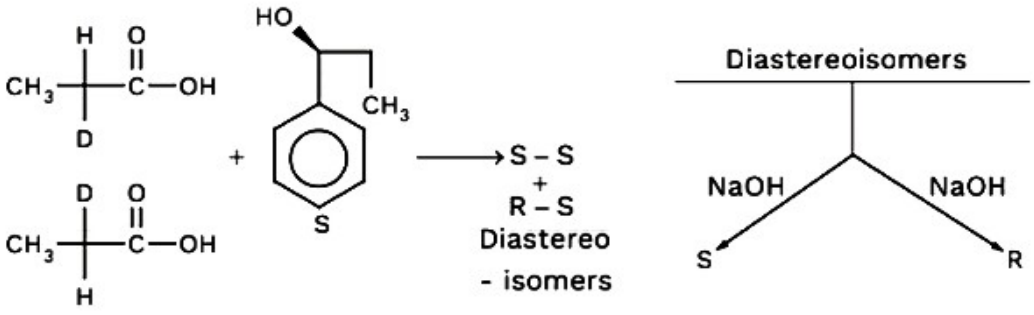
Ex-2



Ex-3



Esterification is used for separation of Enantiomer



(3) Reaction of $-\overset{\overset{O}{\|}}{C}-$ group of Carboxylic Acid :

i) Reduction : $R-\overset{\overset{O}{\|}}{C}-OH + 4H \xrightarrow{LiAlH_4} R - CH_2OH$

NOTE : In the above reaction $-\overset{\overset{O}{\|}}{C}-$ group converts into – CH_2 – group.

(4) Reaction of – COOH group of Carboxylic Acid

For this reaction initially we start from salt of carboxylic acid with base.

i) Alkane Preparation :

$$R-\overset{\overset{O}{\|}}{C}-O\boxed{-H + H-O}-Na \longrightarrow R-\overset{\overset{O}{\|}}{C}-ONa + H_2O$$

$$R-\overset{\overset{O}{\|}}{C}-O\boxed{-H + H-O}-K \longrightarrow R-\overset{\overset{O}{\|}}{C}-OK + H_2O$$

$$R-\overset{\overset{O}{\|}}{C}-ONa \xrightarrow{NaOH-CaO} \underset{\text{Alkane}}{R-H}$$

$$R-\overset{\overset{O}{\|}}{C}-OK \xrightarrow[\text{Kolbe's reaction}]{} \underset{\text{alkane}}{R-R}$$

ii) Alkyl Halide Preparation: -

$$R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{AgOH} R-\overset{\overset{O}{\|}}{C}-OAg \xrightarrow{Br_2/CCl_4} R-Br$$

iii) Carbonyl Compound Preparation (Pyrolysis of calcium salts) :

$$R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{Ca} \boxed{R-\overset{\overset{O}{\|}}{C}-O-Ca-O-\overset{\overset{O}{\|}}{C}-R} \xrightarrow[\Delta]{\text{dry distillation}} R-\overset{\overset{O}{\|}}{C}-R + CaCO_3$$

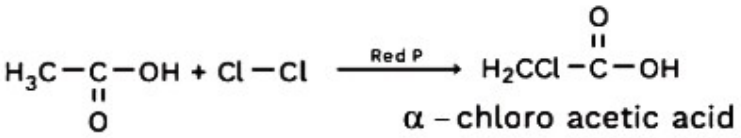
iv) Alkyl Cyanide Preparation:

$$R-\overset{\overset{O}{\|}}{C}-OH \xrightarrow{\ddot{N}H_3} R-\overset{\overset{O}{\|}}{C}-O^{\ominus}\overset{\oplus}{N}H_4 \xrightarrow{\Delta} R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow{P_2O_5,\Delta} R-C\equiv N$$

(5) Reaction of Alkyl group of Carboxylic Acid

i) Hell-Volhard-Zelinsky Reaction :-

Phosphorus reacts with bromine to give phosphorus tribromide, and in the first step this converts the carboxylic acid into an acyl bromide. An acyl bromide can readily exist in the enol form, and this tautomer is rapidly brominated at the α-carbon.



α – chloro acetic acid

NOte : Formic acid does not show the above reaction due to absence of alkyl group.

Special Features of Formic Acid

(a) Formic acid shows acidic character due to the presence of –COOH group as well as shows reducing character due to the presence of –CHO group.



–COOH group –CHO group

(b) It reduces Tollen's reagent and forms silver mirror.

(c) It reduces mercuric chloride and forms black ppt. of mercury.

(d) It decolourises the pink colour if acidic $KMnO_4$.

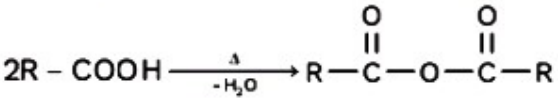
(e) It forms brown ppt. of MnO_2 with basis $KMnO_4$.

(f) It converts the orange colour of acidic $K_2Cr_2O_7$ into green colour.

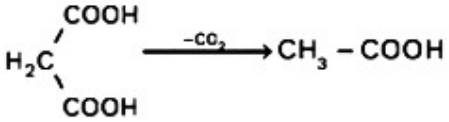
(g) It reduces Fehling solution but benzaldehyde cannot reduce Fehling solution.

Heating Effects Carboxylic Acid

(a) Heating effect on monocarboxylic acid :

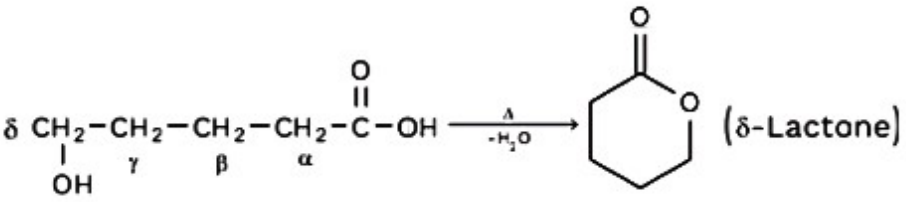


(b) Heating effect on dicarboxylic acid :

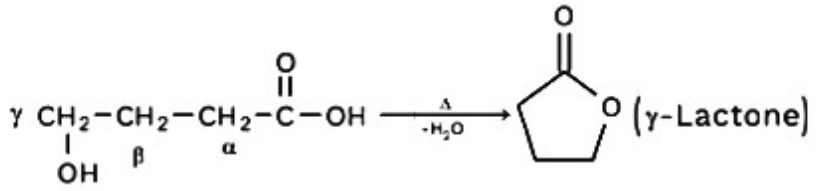


(c) Heating effect on Hydroxy acids :

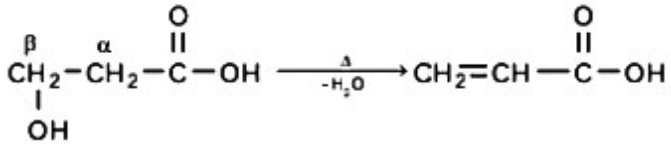
(1) δ-Hydroxy acid



(2) γ-Hydroxy acid

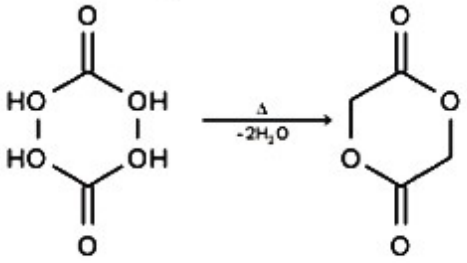


(3) β-Hydroxy acid



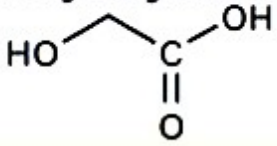
Since 4 or 8 membered rings are less stable the refore β-Hydroxy acids on heating produce unsaturated carboxylic acid.

(4) α-Hydroxy acid

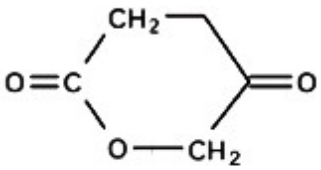


Q34

α-Hydroxy Acid

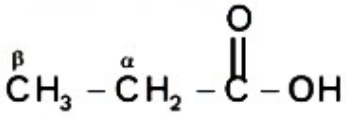


Sol:

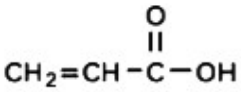


2 – Lactone

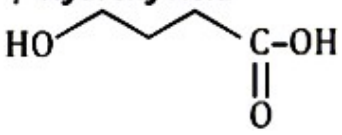
Q35 **β-Hydroxy Acid**



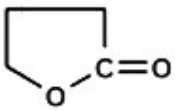
Sol:



Q36 **γ-Hydroxy Acid**



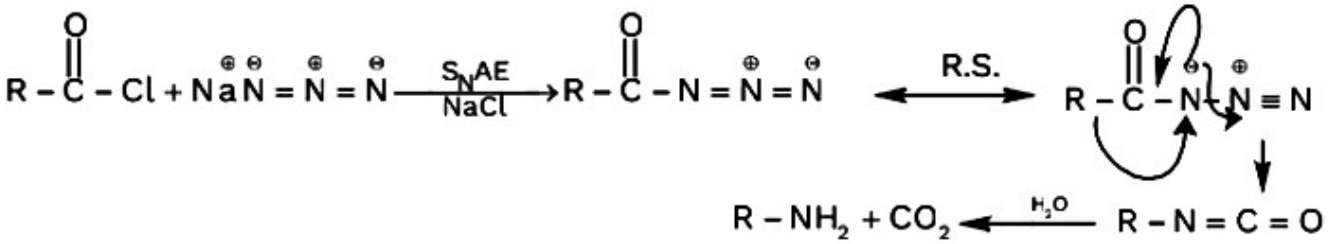
Sol:



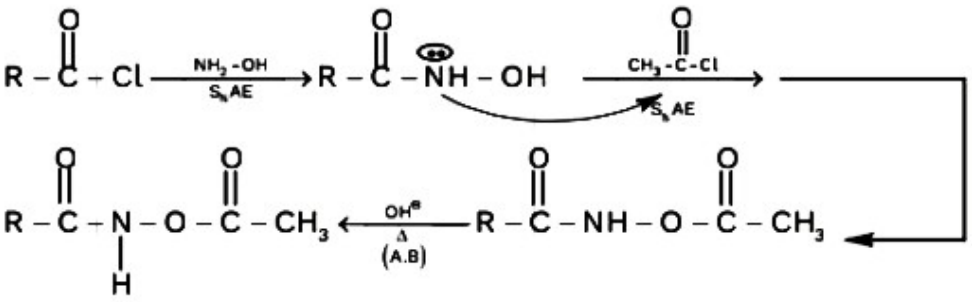
X – lactone

Important Reactions for JEE Advance

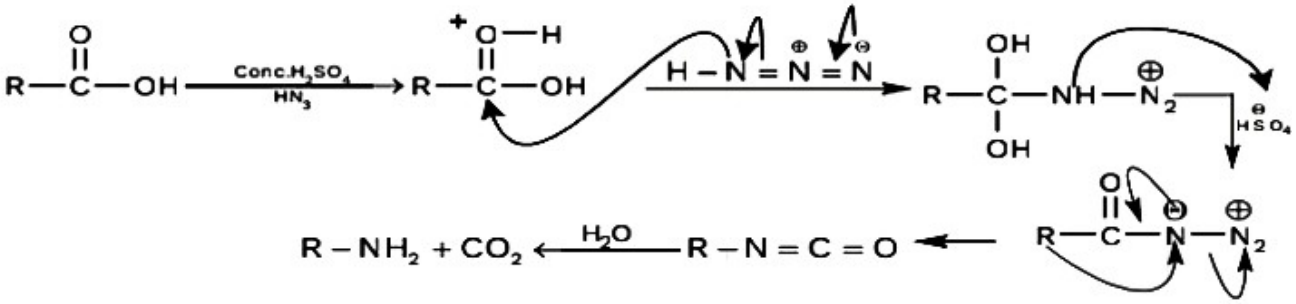
1. Curtius rearrangement:



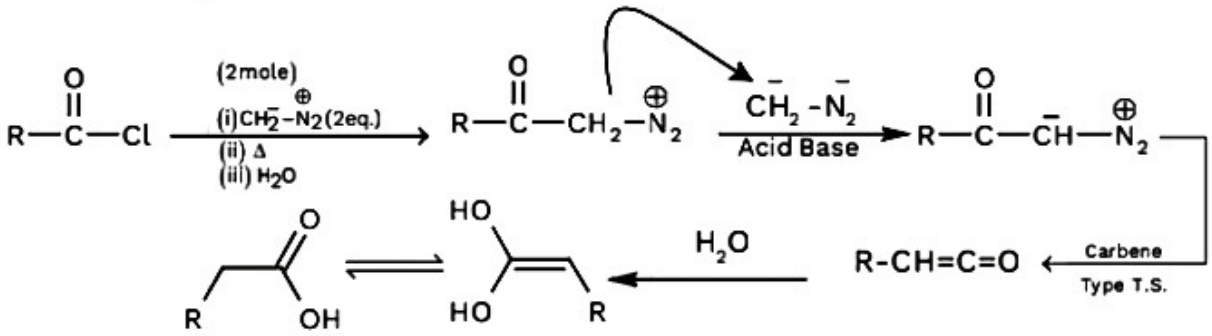
2. Lossen Reaction:



3. Schmidt Reaction:

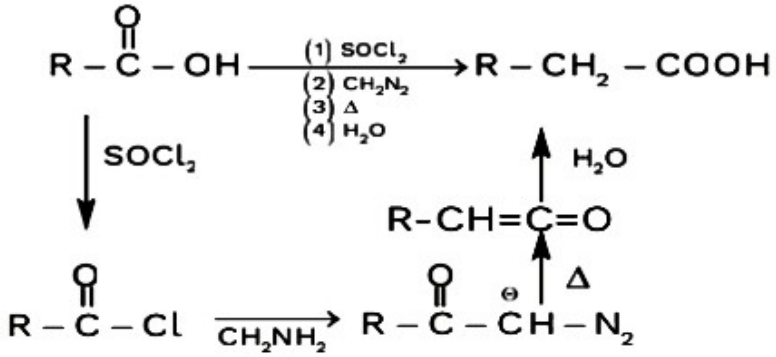


4. Wolf Rearrangement :



5. Aerndt - Eistert reaction

To produce Homologous series



Q37 **Which of the following will give 3-pentanone?**

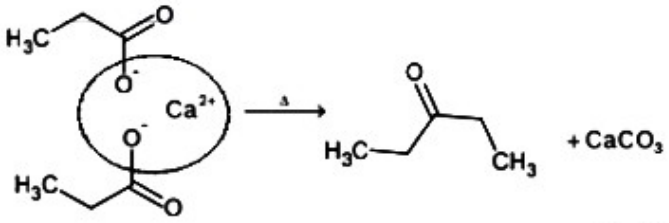
(A) $(CH_3-CH_2COO)_2Ca \xrightarrow{\Delta}$

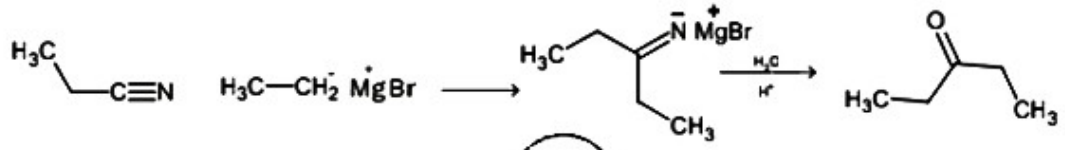
(B) $CH_3-CH_2-C\equiv N \xrightarrow[(ii)\ H_3O^+]{(i)\ CH_3-CH_2-MgBr}$

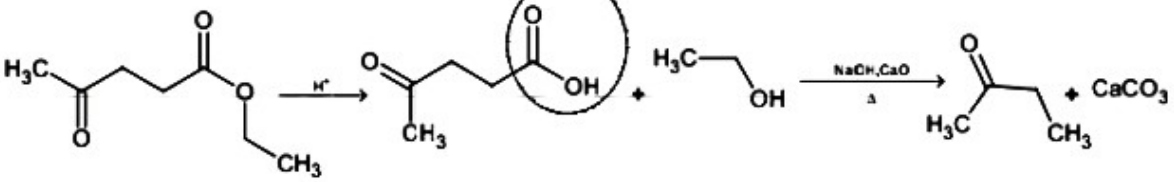
(C) $H_3C-\overset{O}{\overset{\|}{C}}-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OC_2H_5 \xrightarrow[(ii)\ NaOH(CaO)/\Delta]{(i)\ H_3O/H^+}$

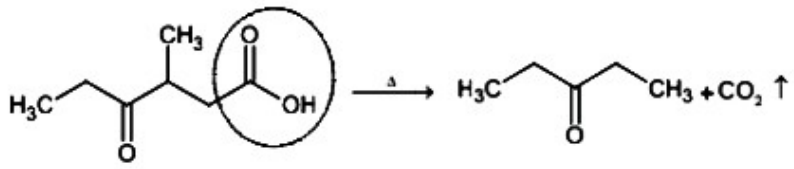
(D) $H_3C-CH_2-\overset{O}{\overset{\|}{C}}-\underset{CH_3}{CH}-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{\Delta}$

Sol:

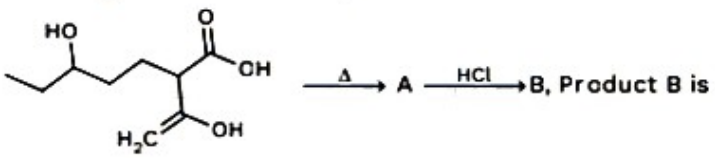
(A) 

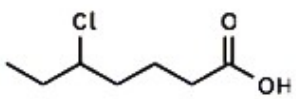
(B) 

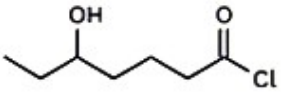
(C) 

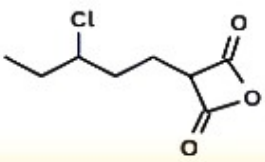
(D) 

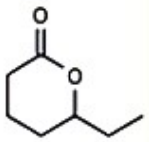
Q38 **In the give reaction sequence:**



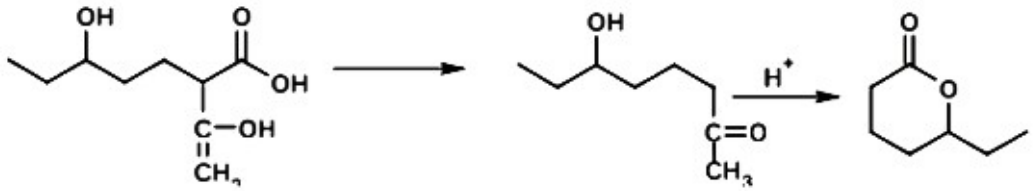
(A) 

(B) 

(C) 

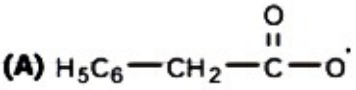
(D) 

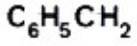
Sol: D



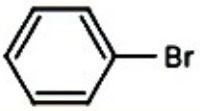
Q39 **The intermediates formed during the following reaction is/are:**

$C_6H_5CH_2COOAg + Br_2 \xrightarrow[\Delta]{CCl_4} C_6H_5CH_2Br$

(A) 

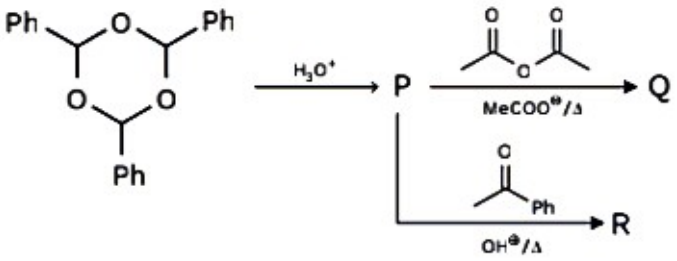
(B) 

(C) $Br^{\cdot}$

(D) 

Sol: **ABC**

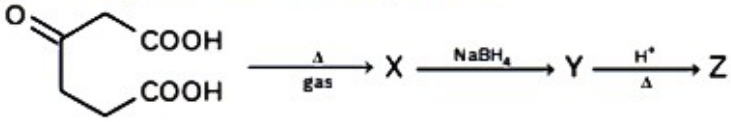
Q40 **Correct statement(s) for given sequence is(are) :**



(A) Q is deactivating in nature but gives o/p directing product
(B) R can be obtained by rearrangement reaction of Ph–CH(OH)–C≡C–Ph
(C) When P is reacted with aq. NaOH, Δ then it will show redox reaction
(D) Q on reaction with H_2/Ni followed by H_3PO_4, Δ produces aromatic compound

Sol: **ABC**

Q41 For the given reaction sequence :



Select the correct statement(s) :

(A) Z is resolvable sweet smelling compound

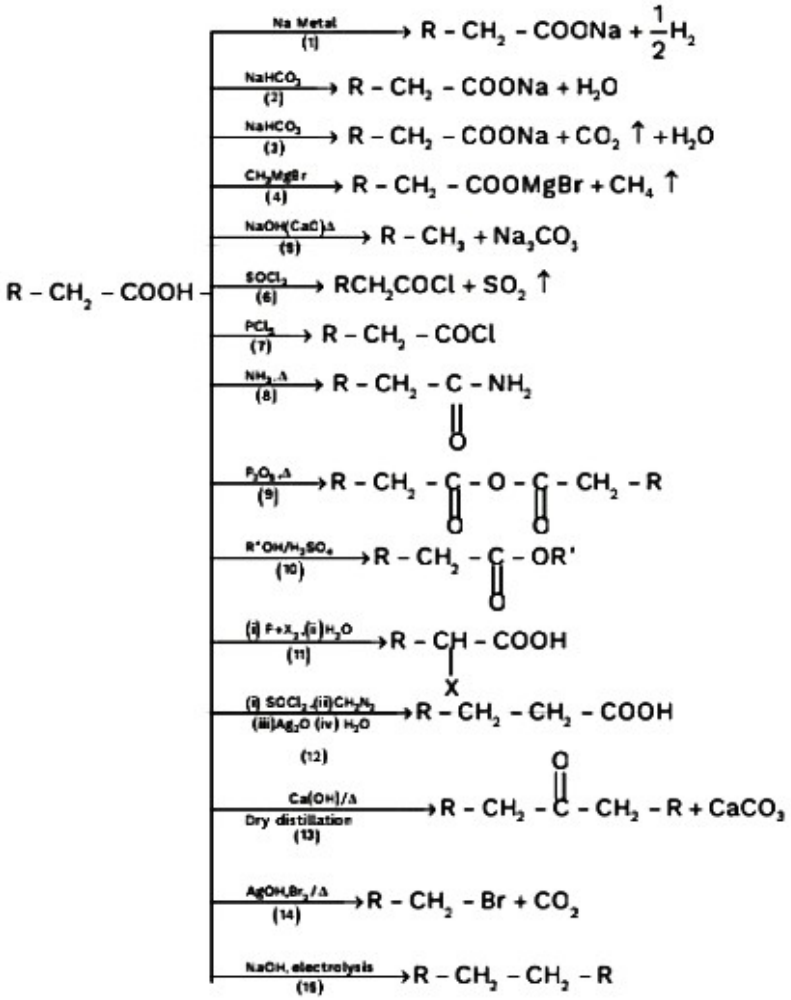
(B) X on reaction with NaOI produces monomer of Nylon-4

(C) Y liberates CO_2 gas with aqueous $NaHCO_3$ and gives yellow crystal with NaOBr

(D) All X, Y and Z are soluble in hot aq. NaOH

Sol: **ACD**

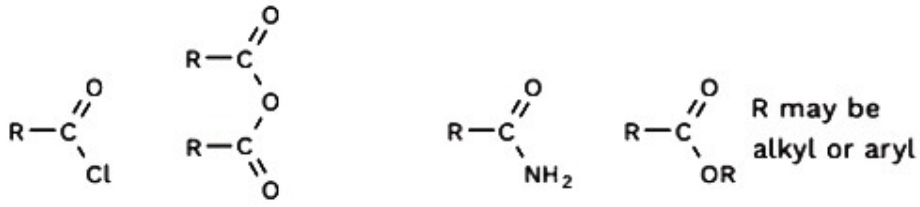
Summary of reactions of carboxylic acids :



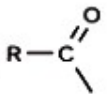
Acid Derivatives (Acid Halide, Ester, Anhydride & Amide):

Closely related to the carboxylic acids and to each other are a number of chemical families known as functional derivative of carboxylic acidic

Acid chlorides, anhydrides, amides and esters. These derivatives are compounds in which the –OH of a carboxyl group has been replaced by –Cl, –OOCR, $-NH_2$, or – OR respectively.



Acid chloride Anhydride Amide Ester

They all contain the acyl group, 

Like the acid to which it is related, an acid derivative may be aliphatic or aromatic, substituted or unsubstituted; whatever the structure of the rest of the molecule, the properties of the function group remain essentially the same.

Condition for acyl nucleophilic substitution reaction:



(i) L must be better leaving group than Nu, i.e, basicity of Nu should be more than that of L

(ii) Nu^- must be a strong enough nucleophilic to attack RCOL.

(iii) Carbonyl carbon must be enough electrophilic to react with $\overset{\ominus}{Nu}$.

Reactivity order:



(A) Acid halides:

Methods of preparation fo Acyl halides:

(i) $RCOOH + PCl_5 \longrightarrow RCOCl + POCl_3 + HCl$

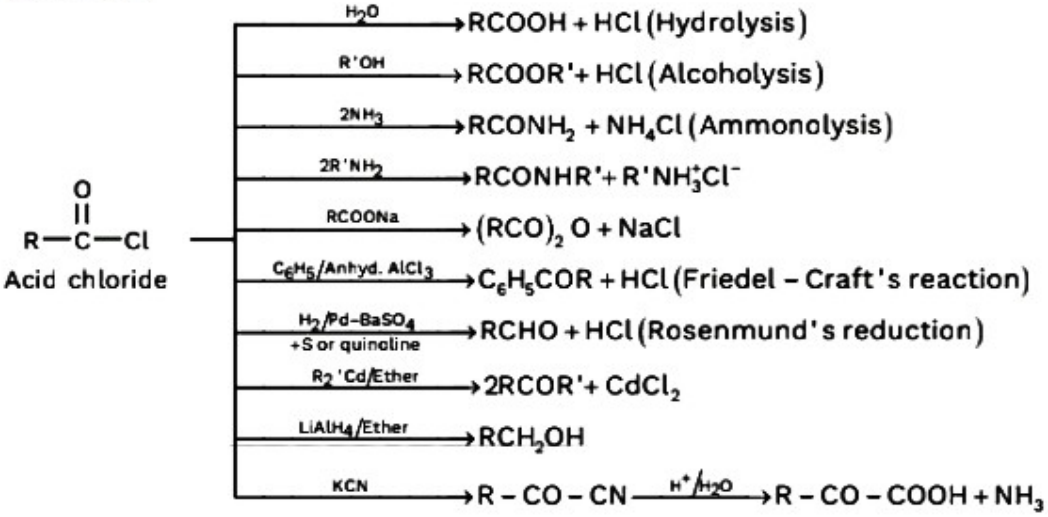
(ii) $3RCOOH + PCl_3 \longrightarrow 3RCOCl + H_3PO_3$

(iii) $RCOOH + SOCl_2 \xrightarrow{\text{Pyridine}} RCOCl + SO_2 + HCl$

Ex. $3CH_3COONa + PCl_3 \xrightarrow{\text{Distil}} \underset{\text{Acetyl chloride}}{3CH_3COCl} + Na_3PO_3$

Ex. $\underset{\text{Sod. benzoate}}{2C_6C_5COONa} + POCl_3 \xrightarrow{\text{Distil}} \underset{\text{Benzoyl chloride}}{2C_6H_5COCl} + NaCl + NaPO_3$

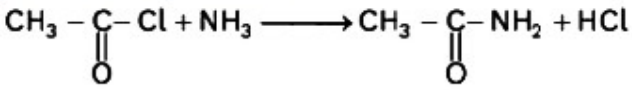
Chemical Reactions:



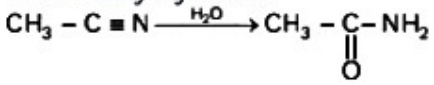
(B) Acid Amide:

Method of prepration :

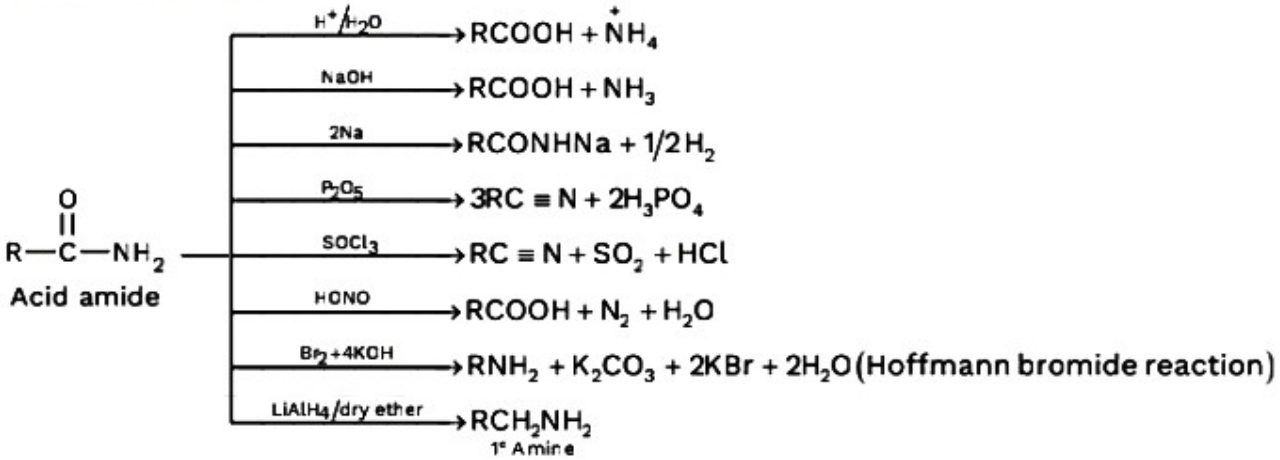
i) From acetychloride



ii) From alkylcyanide



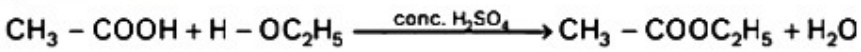
Chemical Reactions:



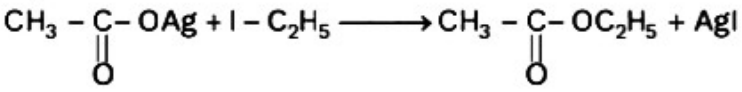
(C) Ester

Method of preprations

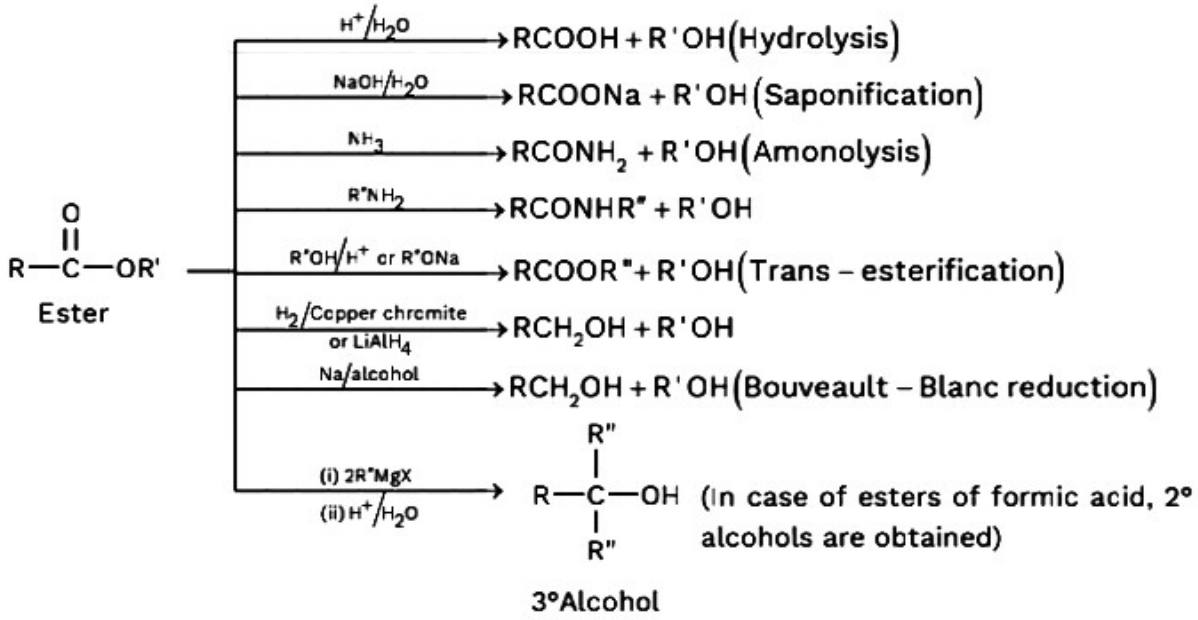
i) By esterification



ii) By silver ethanoate



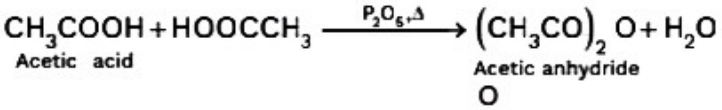
Chemical Reactions:



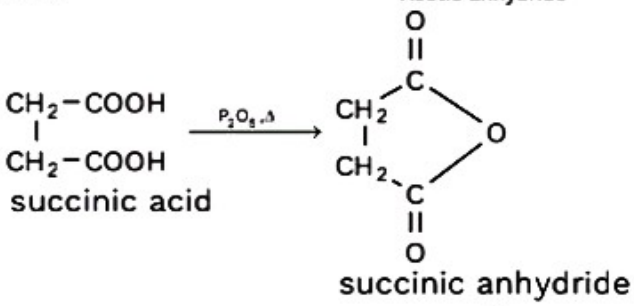
(D) Acid anhydrides :

Methods of Preparation of acid anhydrides :

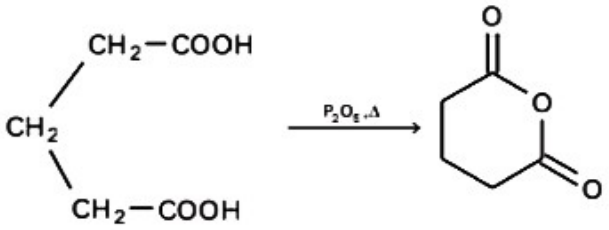
(1) From carboxylic acids :



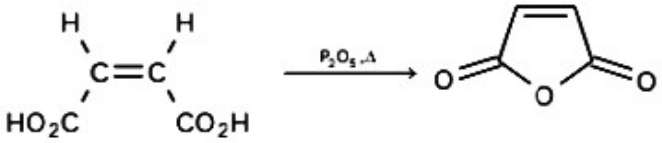
Ex.



Ex.



Ex.



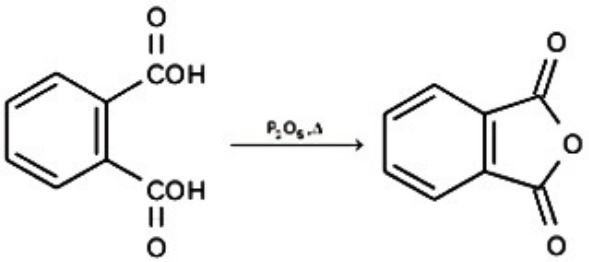
Maleic acid

Maleic anhydride (89%)

+ H_2O

Water

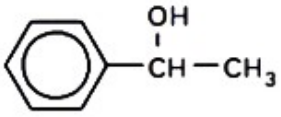
Ex.

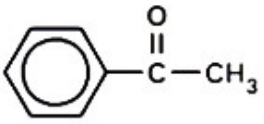


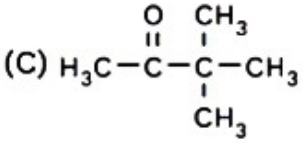
Phthalic acid

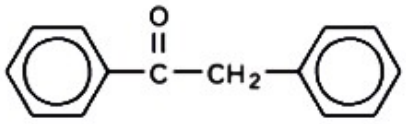
Phthalic anhydride

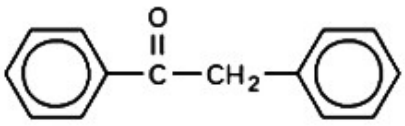
Q42 **Which of the following will not give yellow coloration with NaOI ?**

(A) 

(B) 

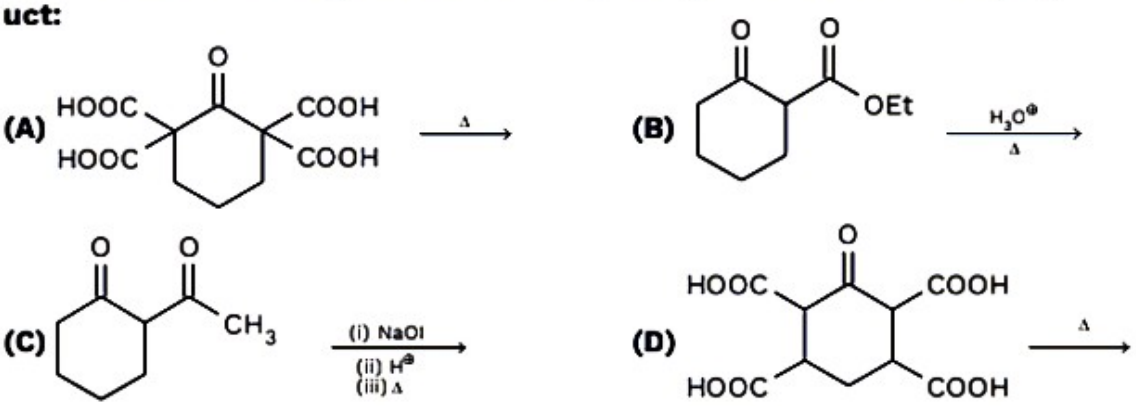
(C) 

(D) 

Sol: 

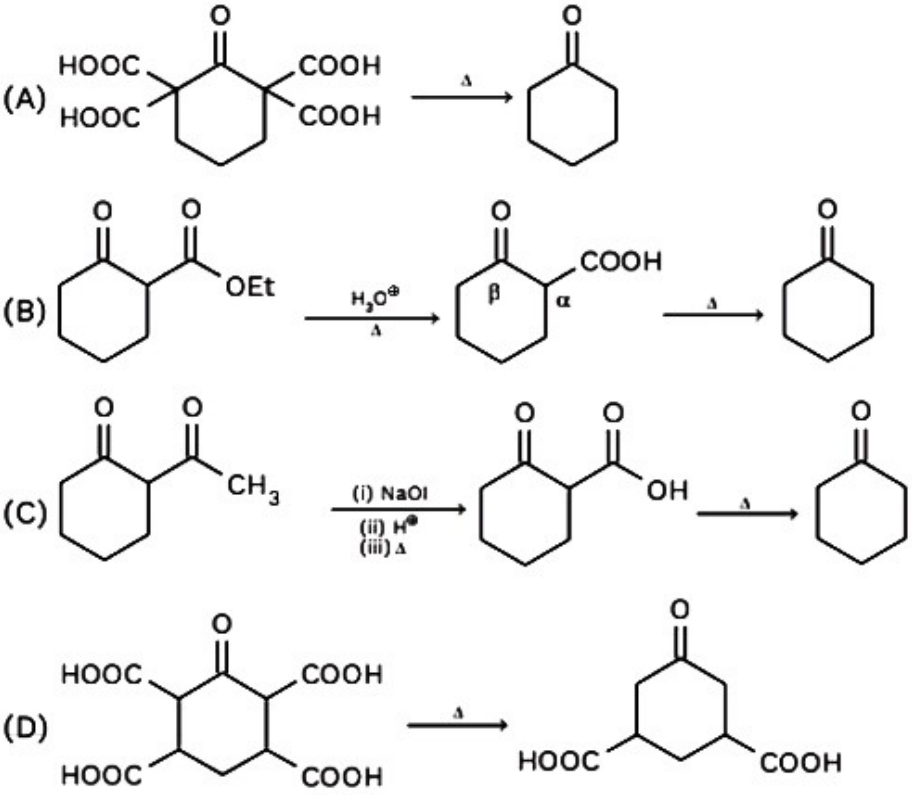
is not a methyl ketone. Thus do not give yellow coloration with NaOI.

Q43 **Which of the following reaction do not yield cyclohexanone as major product:**

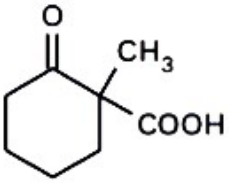


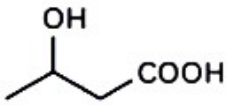
Sol: **(D)**

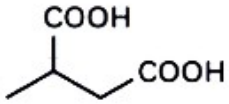
Gem-dicarboxylic and β-keto acids decarboxylates on heating.

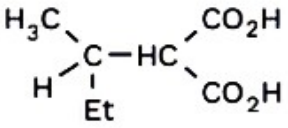


Q44 **Which of the following losses chirality when heated ?**

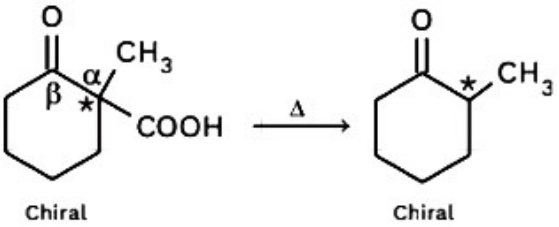
(A) 

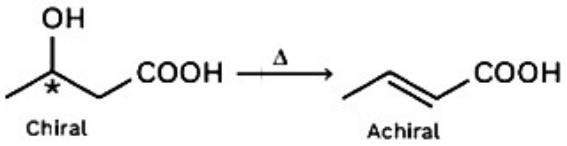
(B) 

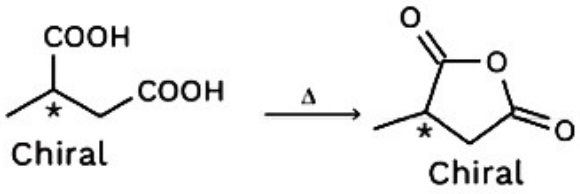
(C) 

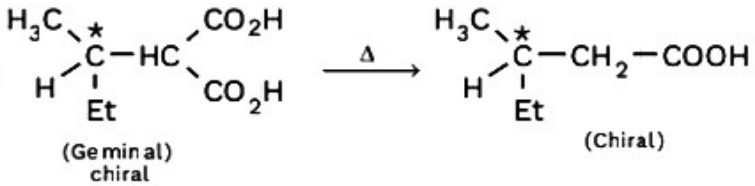
(D) 

Sol: **(B)**

(A) 

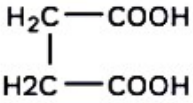
(B) 

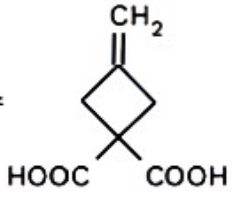
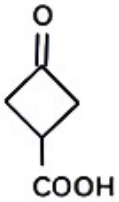
(C) 

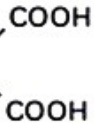
(D) 

Q45 **Which carboxylic acid (A), NE = 52g/eq, loses CO_2 when heated to given an acid (B), NE = 60 g/eq.** $\left(\text{Where NE} = \frac{\text{molecular mass}}{\text{number of ionisable H}}\right)$

(A) A = $CH_2(COOH)_2$; B = CH_3COOH

(B) A =  ; B = CH_3COOH

(C) A =  ; B = 

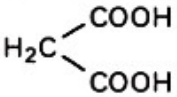
(D) A = A =  ; B = CH_3-CH_2-COOH

Sol: **(A)**

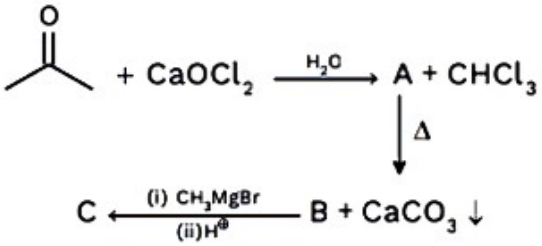
$$NE = \frac{\text{mol. mass}}{H^+}$$

Assume B is monocarboxylic acid and NE = MW the –COOH accounts for 45 g/mol of molecular wt, leaving 60-45 = 15 g/mol for the remainder (due to CH_3–) so "B" is CH_3COOH.

"A" has a second –COOH replacing an H of CH_3COOH, it is malonic acid

 $= \frac{104}{2} = 52g / mol.$

Q46

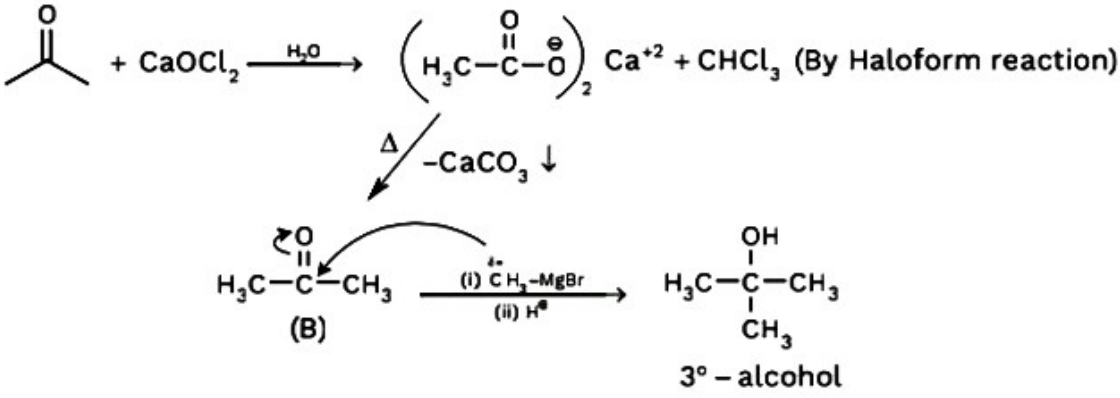


Product is :

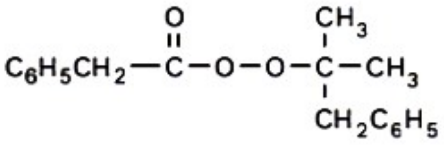
(A) Primary alcohol **(B) Secondary alcohol**

(C) Tertiary alcohol **(D) Phenol**

Sol: **(C)**



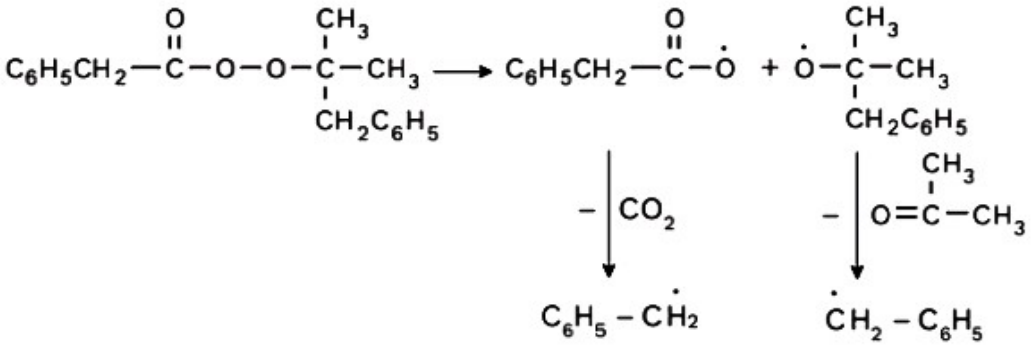
Q47 **What is/are main product/s obtained on thermal decomposition of peroxyester**



(A) $C_6H_5 - CH_2CO\dot{O}$ (B) $H_3C-C(=O)-CH_3$

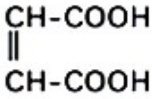
(C) $C_6H_5 - \dot{C}H_2$ (D) $H_3C-C(=O)-CH_2-C_6H_5$

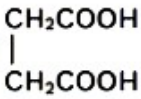
Sol: **(A,B,C)**

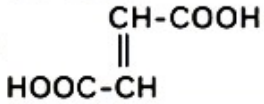


Q48 **Which of the following acid does not form anhydride –**

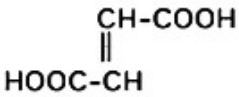
(A) CH_3COOH

(B) 

(C) 

(D) 

Sol: **[D]**



does not form anhydride

Q49 **Formic acid and formaldehyde can be distinguished by treating with –**

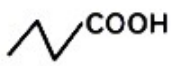
(A) Benedict's solution **(B) Tollen's reagent**

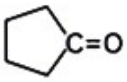
(C) Fehling's solution **(D) $NaHCO_3$**

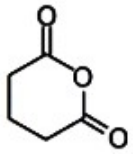
Sol: **[D]**

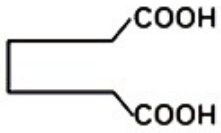
Formic acid is strong acid than formaldehyde formic acid react with $NaHCO_3$ and liberated CO_2 but formaldehyde does not react with $NaHCO_3$.
Therefore option (D) is correct.

Q50 **The product formed when adipic acid is heated–**

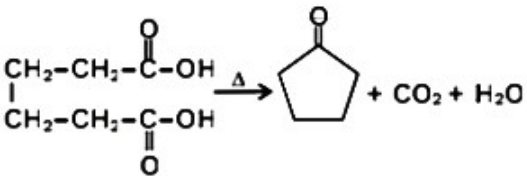
(A) 

(B) 

(C) 

(D) 

Sol: **[B]**



Adipic acid Cyclo pentanone

Adipic acid undergo dehydration as well as decarboxylation to give cyclopentanone when its heated.

So option (B) is correct.

Q51 **Ethyl acetate react with hydrazine to form -**

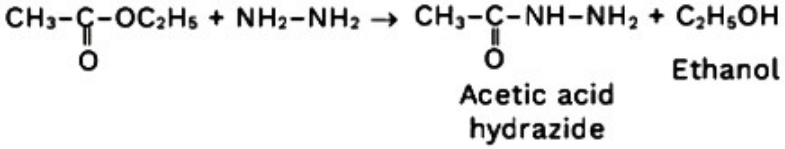
(A) Acetamide and urea

(B) Acetic acid hydrazide and ethanol

(C) Hydroxamic acid and ethanol

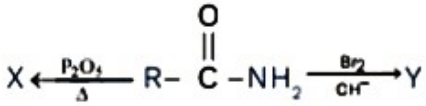
(D) Ethyl isocyanate

Sol: **[B]**



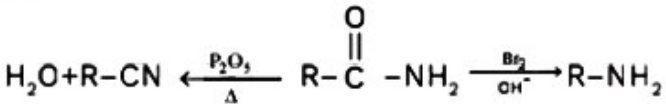
Therefore option (B) is correct.

Q52 **X and Y in the following reaction sequence are–**



(A) R–CN, RCH_2NH_2
(B) RNH_2, RCN
(C) RCN, RNH_2
(D) RCN, RCH_2NH_2

Sol: **[C]**

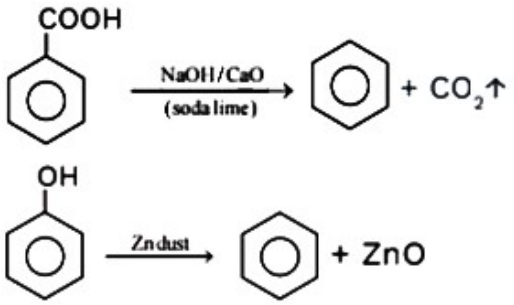


[X] [Y]

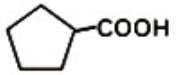
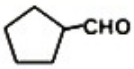
Therefore option (C) is correct.

Q53 **Benzoic acid gives benzene on being heated with X and phenol gives benzene on being heated with Y. Therefore X and Y are respectively –**
(A) Soda lime and copper
(B) Zinc dust and sodium hydroxide
(C) Zinc dust and soda lime
(D) Soda lime and zinc dust

Sol: **[D]**

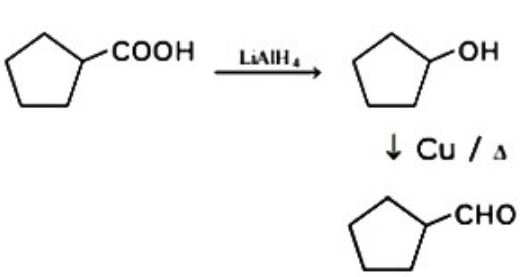


Therefore X and Y are respectively sodalime Zn-dust.
Option (D) is correct.

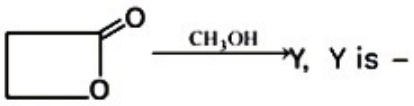
Q54  **is converted into**  **by–**

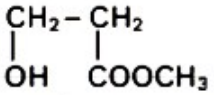
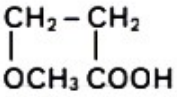
(A) (i) $LiAlH_4$ (ii) Cu/Δ
(B) (i) Cu/Δ (ii) $LiAlH_4$
(C) (i) Ag_2O (ii) Cu/Δ
(D) (i)$SOCl_2$(ii)$LiAl_H4$

Sol: **[A]**

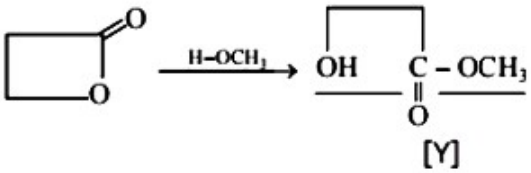


Therefore option (A) is correct

Q55  Y, Y is –

(A) 
(B) 
(C) Both are correct
(D) None is correct

Sol: **[A]**



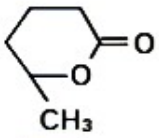
[Y]

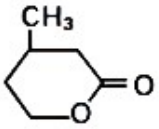
Therefore option (A) is correct.

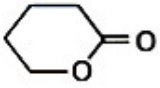
Q56 **End product of this conversion**

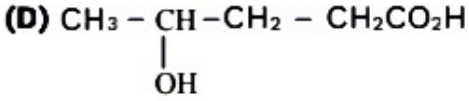


is –

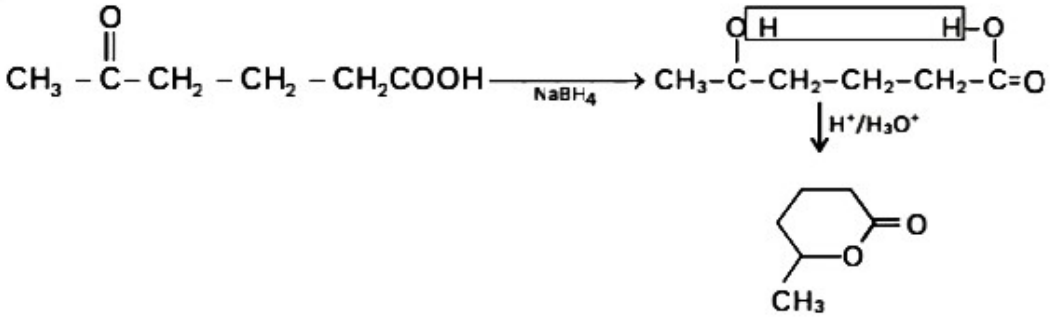
(A) 

(B) 

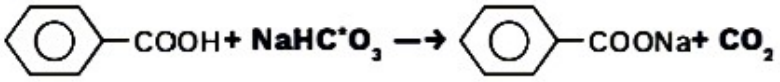
(C) 

(D) 

Sol: **[A]**

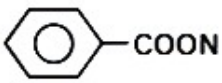


Therefore option (A) is correct.

Q57 

C* is with in the product –

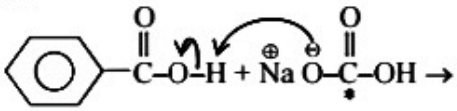
(A) CO_2

(B) 

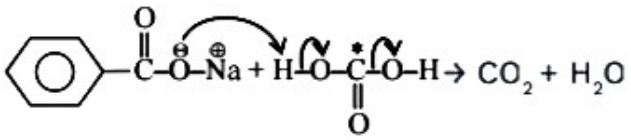
(C) Both

(D) None

Sol: **(A)**

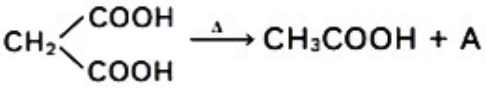


Sod. Bicarbonate



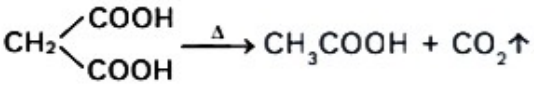
Therefore option (A) is correct.

Q58 **Identify the product A in the following reaction**



(A) CO_2 **(B) CH_3CHO**
(C) CH_3OH **(D) None of these**

Sol: **[A]**



Therefore option (A) is correct.

Q59 **A carboxylic acid (x) & (y) on heating with P_2O_5 (x) give corresponding anhydride but (y) remain unaffected. Carboxylic acid (y) is-**

(A) CH_3COOH **(B) HCOOH**
(C) both (A & B) **(D) CH_3CH_2COOH**

Sol: **[B]**
Carboxylic acid (y) is HCOOH

Q60 **Malonic acid and succinic acid are distinguished by -**

(A) Heating **(B) $NaHCO_3$**
(C) Both A & B **(D) None of these**

Sol: **[A]**
By Heating

Q61 **Which of the following statement is correct -**

(A) Nucleophilic additions to the carbon-oxygen double bond is a characteristic reaction of aldehyde and ketones.

(B) Carboxylic acid and their derivative are characterized by nucleophilic substitution reaction.

(C) In both case initial step involves a nucleophilic attack on the carbonyl carbon but in second step aldehyde and ketone accept a proton to yield additional product whereas in the case of acyl compound leaving group is ejected to form substitutional product.

(D) All are correct

Sol: **[D]**

Factual

Q62 **The relative order of reactivity of acyl derivatives is -**

(A) $R-\overset{\overset{O}{\|}}{C}-Cl > R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > R-\overset{\overset{O}{\|}}{C}-NH_2 > R-\overset{\overset{O}{\|}}{C}-OR'$

(B) $R-\overset{\overset{O}{\|}}{C}-Cl > R-\overset{\overset{O}{\|}}{C}-OR' > R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > R-\overset{\overset{O}{\|}}{C}-NH_2$

(C) $R-\overset{\overset{O}{\|}}{C}-Cl > R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > R-\overset{\overset{O}{\|}}{C}-OR' > R-\overset{\overset{O}{\|}}{C}-NH_2$

(D) None of the above

Sol: **[C]**

$$R-\overset{\overset{O}{\|}}{C}-Cl > R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > R-\overset{\overset{O}{\|}}{C}-OR' > R-\overset{\overset{O}{\|}}{C}-NH_2$$

Q63 **What is the correct order of alkaline hydrolysis of different ester CH_3COOCH_3 [rate = r_1] , $CH_3COOC_2H_5$ [rate = r_2] $CH_3COOC_3H_7$ [rate = r_3] -**

(A) $r_1 > r_2 > r_3$ **(B)** $r_1 < r_2 < r_3$

(C) $r_1 < r_2 > r_3$ **(D)** $r_1 > r_2 < r_3$

Sol: **[A]**

$r_1 > r_2 > r_3$

Q64 **Which of the following esters cannot undergo claisen self condensation**

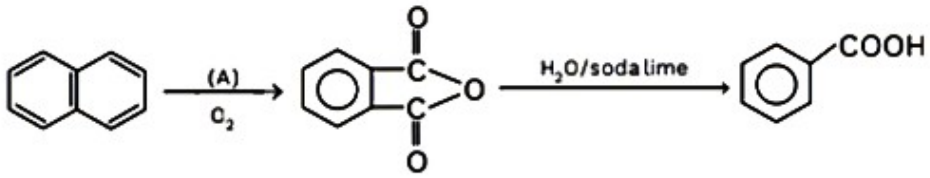
(A) $CH_3CH_2CH_2CH_2COOC_2H_5$ **(B)** $C_6H_5COOC_2H_5$

(C) $C_6H_{11}CH_2COOC_2H_5$ **(D)** $C_6H_5CH_2COOC_2H_5$

Sol: **[B]**

$C_6H_5COOC_2H_5$ esters cannot undergo claisen self condensation

Q65



Oxidizing agent (A) used is

(A) $K_2Cr_2O_7$ / H+ **(B)** AlK . $KMnO_4$

(C) Chromic Acid **(D)** V_2O_5

Sol: **[D]**

Oxidizing agent (A) used is V_2O_5

Q66 **In the reaction**

$$C_6H_5NH_2 \xrightarrow[0-5^oC]{NaNO_2/HCl} (A) \xrightarrow[KCN]{CuCN} (B) \xrightarrow{H^+/H_2O} (C)$$

the product (C) is -

(A) $C_6H_5CH_2NH_2$ **(B)** C_6H_5COOH

(C) C_6H_5OH **(D) None of these**

Sol: **[B]**

Product (C) is C_6H_5COOH

Q67 **Benzoic acid on treatment with hydrazoic acid (N_3H) in the presence of concentrated sulphuric acid gives**

(A) Benzamide **(B) Sodium benzoate**

(C) Aniline **(D)** $C_6H_5CON_3$

Sol: **[C]**

Factual

Q68 **Number of cross products in the given reaction: (Without considering stereoisomers)**

$$CH_3COOC_2H_5 + C_6H_5-CH_2-COOC_2H_5 \xrightarrow[C_2H_5OH]{C_2H_5ONa}$$

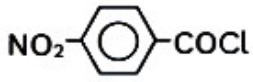
(A) One **(B) Three** **(C) Two** **(D) Four**

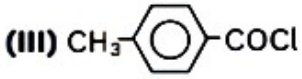
Sol: **[C]**

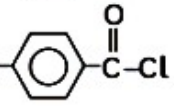
Two

Q69 **Arrange following compounds in decreasing order of reactivity for hydrolysis reactions -**

(I) C_6H_5COCl

(II) 

(III) 

(IV) 

(A) II > IV > I > III

(B) II > IV > III > I

(C) I > II > III > IV

(D) IV > III > II > I

Sol: **[A]**

II > IV > I > III

Q70 **In the given reaction sequence :**

$$CH_3-CH_2-OH \xrightarrow[(ii)H^{\oplus}]{(i)KMnO_4/\overset{\ominus}{O}H/\Delta} (A) \xrightarrow[(ii)NH_3/\Delta]{(i)SOCl_2} (B) \xrightarrow{Br_2/KOH} \textbf{(C) will be}$$

(A) Methylamine

(B) Ethylamine

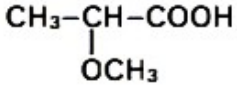
(C) Propylamine

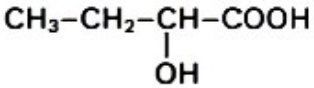
(D) Acetamide

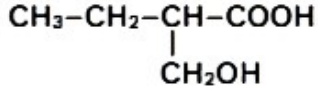
Sol: **[A]**

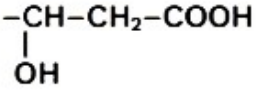
(C) will be Methylamine

Q71 **Which optically active compound on reduction with $LiAlH_4$ will give optically inactive compound ?**

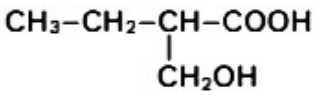
(A) 

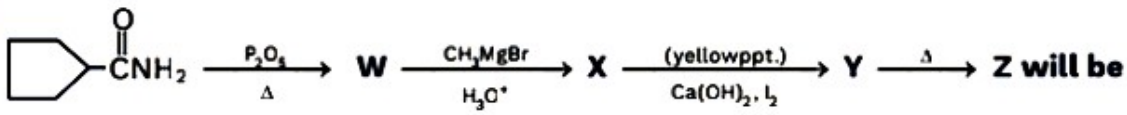
(B) 

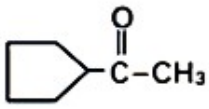
(C) 

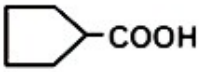
(D) 

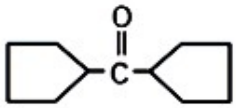
Sol: **[C]**



Q72 

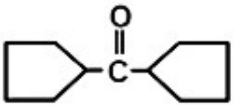
(A) 

(B) 

(C) 

(D) 

Sol: **[C]**

Z will be 

Q73 **Acetonitrile on hydrolysis with a dilute mineral acid gives acetone. True / False**

Sol: $CH_3 - C \equiv N$ (Acetonitrile) $\xrightarrow[\Delta]{HOOH/H^{\oplus}}$ $CH_3COOH + NH_3$ (Acetic acid)

Acetonitrile on hydrolysis with a dilute mineral acid give acetic acid, not an acetone.

Therefore statement is false.

Q74 **Esterfication involves the heating of a carboxylic acid with an alcohol in presence of a protonic acid (H_2SO_4 or HCl gas) as catalyst.**

Sol: In esterfication reaction :

$CH_3-\underset{\underset{O}{\|}}{C}-OH$ (acid) + C_2H_5OH (alcohol) $\underset{\text{or HCl(gas)}}{\overset{H_2SO}{\rightleftharpoons}}$ $CH_3\underset{\underset{O}{\|}}{C}-OC_2H_5$ (ester) + H_2O